

Department of Physics, Princeton University

**Graduate Preliminary Examination
Part I**

Thursday, January 11, 2024
9:00 am - 12:00 noon

Answer TWO out of the THREE questions in Section A (Mechanics) and TWO out of the THREE questions in Section B (Electricity and Magnetism).

Work each problem in a separate examination booklet.

Be sure to label each booklet with your name, the section name, and the problem number.

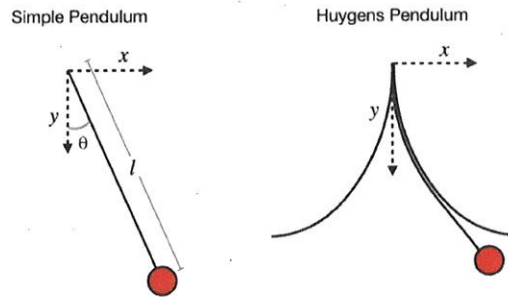
Section A. Mechanics

1. Huygens Pendulum

In the limit of small oscillations, the period of a simple pendulum in a gravitational field, g , is independent of its amplitude (left figure). However, for large values of angle θ the period is a function of the amplitude. Huygens, a Dutch physicist, constructed a pendulum whose period is independent of its amplitude. This was done by allowing the pendulum to wrap around constraining rails that follow the shape of a cycloid (right figure) as it oscillates to the left and right. The x and y Cartesian coordinates of the pendulum's bob are then constrained to follow a trajectory that is parametrized by an angle ϕ (not shown)

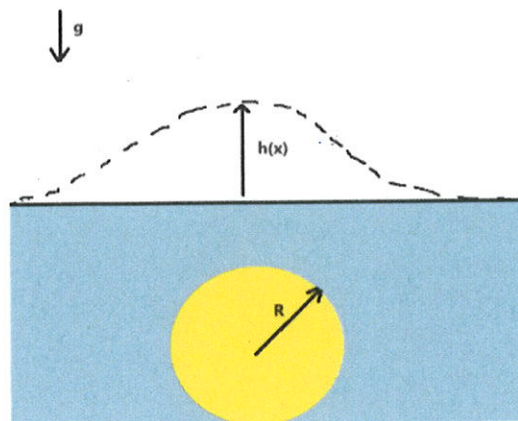
$$x = \frac{l}{4}(\phi + \sin \phi) ; \text{ and } y = \frac{l}{4}(3 + \cos \phi)$$

where l is the length of the string. The equilibrium position of the bob is at $x = 0$, $y = l$, corresponding to $\phi = 0$.



- Show that in the limit of small ϕ the x , y coordinates of the Huygens pendulum have the same values as the x , y coordinates of a simple pendulum with a length l and find the relationship between ϕ and the physical angle θ .
- Derive the Lagrangian of Huygens pendulum. Use ϕ as a generalized coordinate.
- Rewrite the Lagrangian $\mathcal{L}$ in terms of another generalized coordinate u and $\dot{u}$, where $u = \sin(\phi/2)$, and show that the Lagrangian is quadratic in u and $\dot{u}$.
- The motion of Huygens pendulum is periodic in ϕ . Use the Lagrangian in (c) to find the general solution for u as a function of time. Determine the period of Huygens pendulum. How does this period compare with the period of small oscillations of the simple pendulum?

2. Hidden Gold



You decided to hide the excess riches of your grad student salary off the Jersey shore in the format of a sphere of gold with density ρ_g and radius R . Given the ocean density $\rho_o \ll \rho_g$ and Earth's gravity is g , find the new shape of the ocean water surface $h(x)$, i.e, the shift of the water surface level because of the sphere's gravitational attraction. Assume that the height of the water surface would be h_o if the sphere was not there, and you can ignore Earth's curvature. Take x to be the horizontal coordinate of the plane, and you can use that the sphere has a weak effect, that is, $GM/(R - h_0)^2 \ll g$.

3. Ball Toss

You want to throw a ball of mass m at the surface of the earth, so $\vec{g}$ is constant, with initial velocity $\vec{v}_o$ such that the distance r from the origin of the throw to the ball always is increasing. What is the maximum angle to the horizontal θ_m with which the ball can be thrown?

Section B. Electricity and Magnetism

1. Energy of a charged shell

Consider two concentric spheres in space with radii a and $b \geq a$. The space between these radii is filled with a uniform charge density ρ .

(a) Calculate the electrostatic energy U_E stored in the charge distribution in terms of the total charge Q and the variable $x = a/b$. Confirm that your answer is correct in the limit $a \ll b$ and $a = b$.

(b) Find the value for x that corresponds to the minimum energy U_E , keeping the total charge constant. What physical scenario does this correspond to?

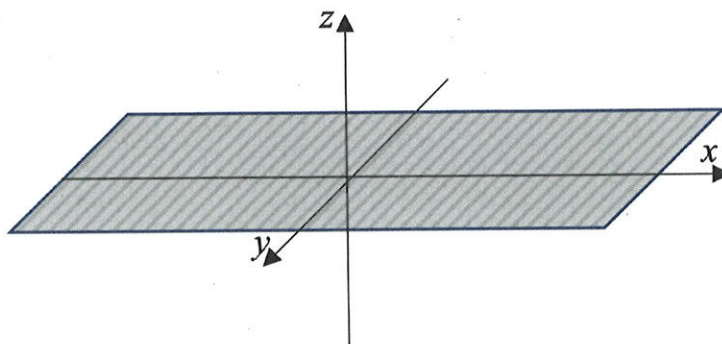
2. An electromagnetic plane wave in a metal

An electromagnetic plane wave of (angular) frequency ω propagates in a metal with conductivity σ , $\epsilon = \epsilon_0$, and $\mu = \mu_0$. Assume that $\sigma \gg \epsilon_0\omega$ (to simplify the algebra, but also because it can be valid for good metals even at optical frequencies.)

- (a) Starting from Maxwell's Equations, find the distance δ over which the amplitude of the electric field is attenuated by a factor of e .
- (b) Find the phase velocity v_p and group velocity v_g of the wave.
- (c) Is v_p greater or less than the speed of light at optical frequencies for metals?

3. Waves in a 2-D electron gas

Consider a classical two-dimensional electron gas confined in the x - y plane at $z = 0$. We will study small-amplitude wave perturbations of the surface charge density of the gas about the equilibrium charge density σ_0 , written in complex notation as $\sigma(x, t) = \sigma_0 + \text{Re}\{Ae^{i(kx - \omega t)}\}$, where $A \ll \sigma_0$. The charge of an electron is $q = -e$ and its mass is m . The frequency of the waves is slow enough that we can ignore electromagnetic radiation and propagation delays and assume quasi-electrostatics throughout the problem.



- Calculate the electric potential associated with the charge perturbation at $t = 0$.
- Calculate the component of the electric field parallel to the electron gas at the interface at any time.
- Write a charge continuity equation applicable to a 2-D electron gas moving in the x direction and simplify it for $A \ll \sigma_0$ to first order in A .
- Obtain a wave equation by calculating $\partial^2 \sigma / \partial t^2$ and find the dispersion relationship for the waves in the form $\omega^2 = f(k)$. What is $f(k)$?

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Graduate Preliminary Examination
Part II

Friday, January 12, 2024
9:00 am - Noon

Answer TWO out of the THREE questions in Section A (Quantum Mechanics) and TWO out of the THREE questions in Section B (Thermodynamics and Statistical Mechanics).

Work each problem in a separate booklet. Be sure to label each booklet with your name, the section name, and the problem number.

Section A. Quantum Mechanics

1. 1D Potential Well

Consider a particle of mass m in one-dimensional (1D) infinite space, with Hamiltonian given by

$$H(x) = -\frac{\hbar^2 \partial_x^2}{2m} + U(x) ,$$

with the potential

$$U(x) = \begin{cases} 0 , & (|x| > L) \\ -V , & (|x| \leq L) \end{cases}$$

where $V > 0$. The eigenstates $\psi(x)$ of the particle satisfy the Schrödinger equation:

$$H(x)\psi(x) = E\psi(x) ,$$

where E is the eigen-energy. Since the potential $U(x)$ is inversion symmetric about $x \rightarrow -x$, the eigenstates have either even or odd parity. Assume the energy $E = \frac{\hbar^2 p^2}{2m} - V$ in terms of wave number $p > 0$.

(a) Solve for the **odd parity** (namely, $\psi(x) = -\psi(-x)$) eigenstates which are **bound states** with energy $E < 0$. Give the expression of the eigenstate wavefunction (normalization is not required), and find the equation that p should satisfy.

(b) Find the lower bound V_1 of V for odd parity bound states to exist for $V > V_1$.

(c) In the limit $V \rightarrow V_1$ ($V > V_1$), for the lowest energy odd parity bound state, compute its eigen-energy E_1 up to leading non-vanishing order in $(V - V_1)$.

2. Harmonic oscillator with resonant forcing

Consider a one-dimensional harmonic oscillator with mass m and frequency ω . The Hamiltonian is

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2$$

For $t \leq 0$ the oscillator is in the ground state. For $0 < t < T$, where $T = \frac{2\pi}{\omega}$ is the period of the oscillator, we turn on a resonant sinusoidal perturbation (the frequency of the perturbation is the same as that of the harmonic oscillator):

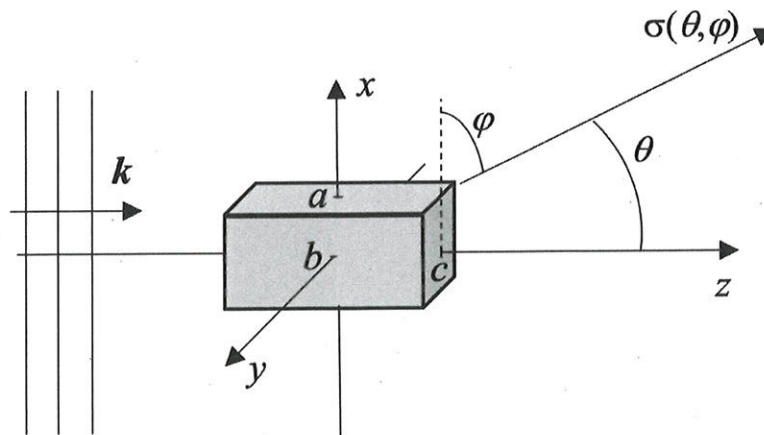
$$V(x, t) = F_0 x \sin \omega t$$

The perturbation is switched off for $t \geq T$.

- (a) Find the state $|\psi(t)\rangle$ for all positive times. Use first order time-dependent perturbation theory throughout this problem.
- (b) What are the conditions so that first order time-dependent perturbation theory gives a good approximation for the state $|\psi(t)\rangle$?
- (c) Compute the expectation value $\langle x \rangle$ of the oscillator's position as a function of time t , both for $0 < t < T$ and $t \geq T$.
- (d) What is the probability that the oscillator is found in the first excited state for $t \geq T$?

3. Scattering from a rectangular box

Consider a 3-dimensional box potential $V(x, y, z) = V_0$ for $-a < x < a$, $-b < y < b$, $-c < z < c$ and $V = 0$ otherwise. In other words, the potential is equal to V_0 inside a box of dimensions $2a$ in x , $2b$ in y and $2c$ in z and is zero everywhere else. A particle of mass m is traveling along the z axis with momentum k .



- Using the Born approximation calculate the differential scattering cross-section $\sigma(\theta, \phi)$ for the scattering of the particle. While the scattering direction is described by spherical coordinates, it may be easier to do the calculation in Cartesian coordinates.
- Consider now the limit of low-energy scattering, so $ka \ll 1$, $kb \ll 1$, $kc \ll 1$. Simplify your cross-section in this limit.
- What happens when the box is very long in one direction, for example, $c \gg b, a$. What is the condition to be able to tell the orientation of the box from the scattering cross-section?

Section B. Statistical Mechanics and Thermodynamics

1. An Ideal Gas Cycle

Consider one mole of an ideal, monatomic gas. We will follow the gas around a thermodynamic cycle in which the gas starts at $P_1 = 1$ atm and $T_1 = 300$ K, is compressed adiabatically to half of its initial volume (to temperature T_2), then is heated at constant volume through a temperature rise of $\Delta T = 1600$ K (to temperature $T_3 = T_2 + 1600$ K), then expands adiabatically back to its initial volume (to temperature T_4), and finally cools back to its initial temperature at fixed volume.

- (a) Determine the unknown temperatures T_2, T_3, T_4 .
- (b) For each leg of the cycle determine the heat absorbed by the gas. What is the net entropy change of the gas around the cycle?
- (c) If the source of the heat is a reservoir at $T_h = 3000$ K and the heat rejection is to a reservoir at $T_l = 200$ K, what is the entropy change of the universe when the cycle is completed?

2. A harmonic oscillator in thermal equilibrium

Consider a simple 1-D harmonic oscillator in thermal contact with and at equilibrium with a thermal reservoir at a temperature T . The energies of the oscillator eigenstates, including zero-point energy, are given by

$$E_n = (n + \frac{1}{2})\hbar\omega.$$

Such a harmonic oscillator can be used as a detector for very rare external events that occasionally deposit energy into the oscillator. One can ask if quantum mechanics places a limit on the detection of such rare events.

a) Calculate the partition function $Z(T)$, the free energy $F(T)$, the entropy $S(T)$, the average energy $E(T)$ and the heat capacity $C_V(T)$ of the oscillator as a function of temperature T .

b) Find fluctuations of the average energy of the harmonic oscillator $\delta E = \sqrt{E^2 - (\overline{E})^2}$. What is the limit of energy fluctuations when the temperature becomes low compared to the energy of the oscillator, $kT < \hbar\omega$?

3. A classical polymer

Consider a polymer with N monomers at positions $\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N$. Assume that the links among monomers are rigid, so that $|\mathbf{x}_n - \mathbf{x}_{n-1}| = r_0$. Further assume that the chain is “freely jointed,” so that the orientation of the vector $\mathbf{r}_n = \mathbf{x}_{n+1} - \mathbf{x}_n$ is random relative to the vector $\mathbf{r}_{n-1}$.

- (a) Show, in the absence of any interactions among the monomers, that as N becomes large the distribution of the end-to-end vector $\mathbf{R} = \mathbf{x}_N - \mathbf{x}_0$ is Gaussian,

$$P(\mathbf{R}) = \left(\frac{1}{\sqrt{2\pi N r_0^2/d}} \right)^d \exp \left[-\frac{|\mathbf{R}|^2}{2N r_0^2/d} \right],$$

where d is the dimensionality of space. Check explicitly that this distribution is normalized.

- (b) What is the distribution, $P(R)$, of the end-to-end length $R = |\mathbf{R}|$? It is important to get the functional form of $P(R)$ but you don't need to worry so much about constants out in front.
- (c) The polymer is in equilibrium with a surrounding solvent at temperature T . This means that the probability $P(R)$ should be related to a free energy $F_0(R)$,

$$P(R) = \frac{1}{Z_0} \exp \left[-\frac{F_0(R)}{k_B T} \right].$$

Find the form of $F_0(R)$ and the distance $R = R_*$, that minimizes F_0 . Note that in computing $F_0(R)$ you can drop terms that are independent of R . You should find that $R_* \propto N^{\nu_0}$. Does the *exponent* ν_0 depend on the dimensionality of space?

- (d) If you pull on the polymer and try to stretch beyond $R = R_*$ you will feel a stiffness. Give an expression from this stiffness, which you can derive from the free energy.